Properties

(i) $F(x)$ is a non decreasing function of x

(ii) $0<F(x)<1 \qquad -\infty < x < \infty$

(iii) $F(-\infty) = 0,$

(iv) $F(\infty) = 1$

(v) For any real constant a and b such that $a < b,\ p(\ a < X \leq b) = F(b) - F(a)$

(vi) $f(x) = \frac{d}{dx}(F(x)) \qquad$ i.e., $f(x) = F'(x)$

Example 9.5

A random variable X has the following probability mass function

X	0	1	2	3	4	5	6
$P(X = x)$	a	$3a$	$5a$	$7a$	$9a$	$11a$	$13a$

(i) Find the value of 'a'

(ii) Find the c.d.f $F(x)$ of X

(iii) Evaluate : (a) $P(X \geq 4)$ (b) $P(X < 5)$ (c) $P(3 \leq X \leq 6)$

(iv) $P(X = 5)$ using $F(x)$

Solution:

(i) Since $P(X = x)$ is probability mass function $\sum P(X = x) = 1$

i.e., $P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6) = 1$

$$a + 3a + 5a + 7a + 9a + 11a + 13a = 1$$

$$49a = 1 \Rightarrow a = \frac{1}{49}$$

(ii) Hence the c.d.f is

X	0	1	2	3	4	5	6
$P(x)$	$\frac{1}{49}$	$\frac{3}{49}$	$\frac{5}{49}$	$\frac{7}{49}$	$\frac{9}{49}$	$\frac{11}{49}$	$\frac{13}{49}$
$F(x)$	$\frac{1}{49}$	$\frac{4}{49}$	$\frac{9}{49}$	$\frac{16}{49}$	$\frac{25}{49}$	$\frac{36}{49}$	$\frac{49}{49} = 1$